

Torque Feedback MTPA Control using an Estimated Torque Approximation Equation

Sota Kawashima^{*a)} Member, Keiichiro Kondo^{*} Senior Member
Kazuhiko Matsunami^{**} Member

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The operation of interior permanent magnet synchronous motors (IPMSMs) can deviate from the optimal point for vector control to magnetic saturation of inductance and variations in the magnet flux caused by changes in the magnet temperature. Torque feedback maximum torque per ampere (MTPA) control, which is known for its resilience to parameter fluctuations. This control method requires accurate torque estimation and MTPA angle estimation. However, conventional methods decreased estimation accuracy inductance or magnet flux a method to enhance the accuracy of these estimations by combining torque approximation equations derived from real motor drives. The accuracy of the proposed method in estimating the torque and MTPA angle, as well as its performance under steady-state torque control and transient conditions, validated.

Keywords: IPMSM, torque feedback control, magnet temperature, magnetic saturation, torque estimation, MTPA control

1. Introduction

In recent years, as global efforts toward carbon neutrality accelerate, there has been increasing attention on electric vehicles (EVs). The competition to develop EVs is intensifying worldwide, requiring the creation of high-performance products within shorter timeframes.

Electric vehicles primarily use Interior Permanent Magnet Synchronous Motors (IPMSMs) for propulsion because of their compact size and high efficiency. However, IPMSMs experience fluctuating inductance values depending on the drive current, and the electromotive force constant varies with the temperature of the permanent magnets in the rotor. These variations in magnetic parameters hinder the controller from determining the minimum required current for a desired torque reference, thereby reducing steady-state torque output performance.

Among these issues, the fluctuation of inductance due to magnetic saturation primarily arises from the current vector. An offline approach exists, involving pre-acquiring the relationship between current and inductance in the form of tables or mathematical models, and subsequently referencing this to adjust the inductance values used in control^{(1)–(6)}. There are methods such as generating current reference values based on a pre-acquired Lookup Table (LUT)⁽¹⁾, combining the LUT for current-torque and flux with a real-time flux estimation mechanism for compensation^{(2)–(4)}, creating a flux

model in advance that considers the fluctuation of magnetic parameters⁽⁵⁾, and combining a pre-created model (equivalent circuit) with a real-time flux estimation mechanism⁽⁶⁾. Conversely, because the magnetomotive force of permanent magnets has temperature dependence, it varies according to the operating conditions. Therefore, online methods^{(7)–(12)}, which compensate for PMSM parameter fluctuations through control calculations using information available during operation, are effective. Among these methods, a technique to estimate magnetic parameters by injecting harmonics has been proposed^{(7)–(9)}, or within a microcontroller using numerical calculation methods^{(10)–(12)}. Generally, offline methods can accurately compensate for parameter fluctuations when the current is used as an argument, but they are time-consuming and labor-intensive. However, in applications such as automobiles, where the production volume per model is high, these methods are particularly useful. Moreover, previous studies (2)–(4)(6), or (10)–(12) have employed methods that combine both offline and online approaches to estimate the real-time magnetic flux status inside the motor for use in control. Therefore, for automotive applications, combining the effectiveness of both offline and online methods proves to be advantageous.

In controlling IPMSMs, Maximum Torque Per Ampere (MTPA) control is employed to minimize copper losses and achieve high-efficiency operation. Determining the magnitude and phase of the current vector is essential for this purpose. Even when magnetic parameters vary, torque feedback MTPA control has been proposed as a method to operate close to the MTPA operating point⁽¹³⁾. In this approach, the calculated value of the IPMSM's torque is obtained by dividing the inverter output power by the rotational speed. However, although the phase angle requires magnetic parameters, calculations were often performed using nominal values,

a) Correspondence to: Sota Kawashima. E-mail: kawa4ma-sota16@fuji.waseda.jp

* Waseda University
3-4-1, Okubo, Shinjuku-ku, Tokyo 169-0051, Japan

** Suzuki Motor Corporation
300, Takatsuka-cho, Chuo-ku, Hamamatsu, Shizuoka 432-8611, Japan

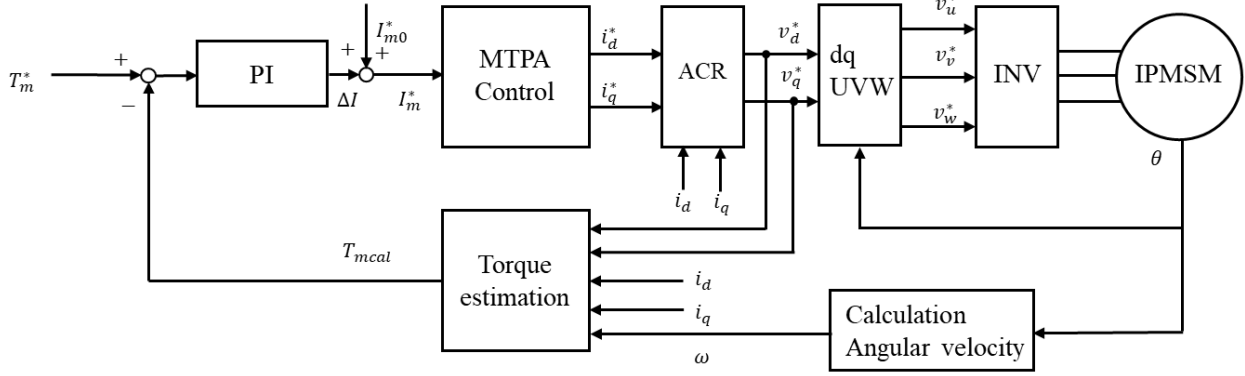


Fig. 1. Block diagram of torque feedback MTPA control

resulting in MTPA control not always being achieved⁽¹⁴⁾.

In this study, torque feedback MTPA control utilizes tables of current vectors versus torque obtained offline for each electromotive force constant of permanent magnets to perform torque and MTPA calculations. Specifically, by referencing the appropriate current vector-torque table from the permanent magnet's flux electromotive force obtained from the voltage reference value during operation, it sets the MTPA phase angle and the feedback value for torque. This proposes a magnetic parameter-free method for steady-state torque control. By acquiring an approximation formula for torque in relation to current and temperature offline and reflecting the information of temperature (permanent magnet electromotive force constant) and current into this torque approximation formula online during control, it is possible to generate current vector references that account for flux variations due to current and temperature changes without using magnetic parameters. Thus, this method can be considered a certain control method that combines the benefits of both the online methods of torque feedback and permanent magnet flux estimation, and the offline method with the inductance table, without requiring complex analyses or special sensors for torque and temperature. This paper verifies the estimation accuracy of torque and MTPA angle, steady-state torque control performance, and transient control performance proposed by the method.

Consequently, the error in the output torque, which is 0.1 Nm order with the conventional method for a torque reference of 1.0 Nm order, can be reduced to 0.01 Nm order with the proposed method.

2. Torque Feedback MTPA Control

2.1 Torque Feedback MTPA Control Figure 1 depicts a block diagram of a typical torque feedback MTPA control system. This system integrates MTPA control (Fig. 3) upstream of the PMSM current control system⁽¹⁵⁾ (Fig. 2), combined with torque feedback control.

In the PMSM current control system, all the effects of the motor's magnetic parameter fluctuations are reflected in the inverter output voltages v_d and v_q .

In the torque feedback MTPA control, the torque feedback value T_{mcal} is calculated using the voltage reference values v_d^* and v_q^* , and the current vector magnitude I_m^* is determined to compensate for the deviation from the torque reference value T_m^* . Subsequently, MTPA control determines the

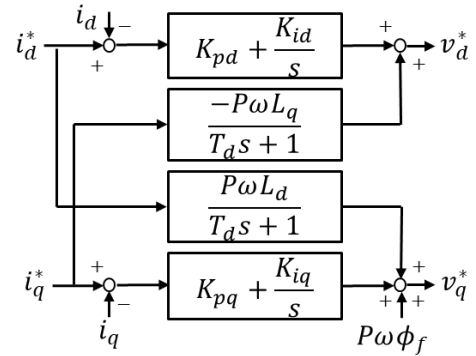


Fig. 2. Block diagram of current control (ACR: Automatic Current Regulator)

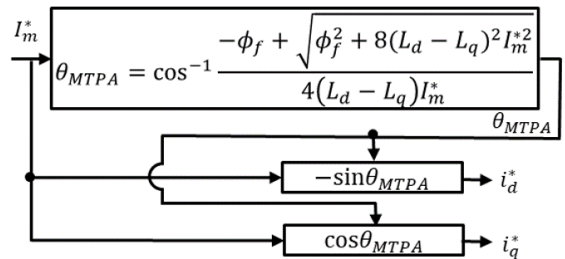


Fig. 3. Block diagram of MTPA control

current phase angle θ_{MTPA} that can produce maximum torque with the set current vector magnitude I_m^* , generating the d-q axis current reference values.

As a result, high-efficiency vector control that minimizes the current required to meet the desired torque reference is achieved.

2.2 Conventional Method and Their Problems To achieve motor operation at the highest efficiency point through torque feedback MTPA control, accurate torque estimation and MTPA angle computation are essential. Traditionally, two methods have been employed for torque estimation.

In this paper, (1) is referred to as the current-type torque estimation formula.

$$T_{mcal} = P\{(L_d - L_q)i_d + \phi_f\}i_q \dots \dots \dots (1)$$

L_d : daxis inductance L_q : qaxis inductance

ϕ_f : Electromotive force constant

The current-type torque estimation formula is unaffected

by errors between controller voltage reference values and inverter output voltage, as it does not use voltage values in the calculation. Therefore, when the motor parameters are correct, accurate torque estimation is achievable across the entire speed range. However, if magnetic parameters are assumed to be fixed values, the accuracy of torque estimation diminishes when they fluctuate.

In this paper, (2) is referred to as the voltage-type torque estimation formula.

$$T_{mcal} = \frac{P \sum_{k=u,v,w} \left\{ v_k^* - \left(\frac{T_{dt}}{T_{sw}} V_{dc} + V_{fd} \right) \text{sign}(i_k) - R i_k \right\} i_k}{\omega} \quad (2)$$

ω : angular velocity T_{dt} : dead time v_k^* : Voltage reference
 T_{sw} : switching cycle V_{dc} : Supply voltage i_k : phase current
 V_{fd} : Voltage drop at element R : Winding resistance

The voltage-type torque estimation formula calculates torque by dividing the active power by the angular velocity. This approach capitalizes on the reflection of motor magnetic flux fluctuations in the inverter output voltage vector. Since magnetic parameters are not involved in the torque calculation, accurate estimation is achievable even during these fluctuations. However, the ratio of losses to motor input power rises in the low-speed range, complicating the determination of the active power contributing to machine output. Additionally, as the active power is divided by the angular velocity, errors are magnified at lower speeds, resulting in decreased accuracy of torque estimation.

In contrast, (3) represents the MTPA theoretical equation, which is used for the calculation of MTPA control. This equation is derived by partially differentiating the torque theoretical equation (1) with respect to the current phase angle θ and solving the extreme value problem.

$$\theta_{MTPA} = \cos^{-1} \frac{-\phi_f + \sqrt{\phi_f^2 + 8(L_d - L_q)^2 I_m^{*2}}}{4(L_d - L_q) I_m^*} \quad (3)$$

The accuracy of calculating the MTPA angle is high when the magnetic parameters are correct. However, this accuracy diminishes when these parameters vary.

3. Algorithm for Creating Torque Estimation Approximation Equations

3.1 Steady-state Torque Estimation Method In this paper, a set of torque estimation approximation equations is developed based on the torque estimated at every operating point of the motor. Accurate torque estimation is crucial for this purpose, but challenges are outlined in Section 2.2. Hence, the authors propose a method aimed at achieving high torque estimation accuracy under steady-state conditions⁽¹⁶⁾. The torque estimation method is as follows.

In an electric motor, torque is represented by (4) as follows:

$$T_{mcal} = P(\phi_d i_q - \phi_q i_d) \quad (4)$$

The flux vector is expressed by (5) as

$$\begin{cases} \phi_q = -\frac{\partial v_d^*}{\partial \omega} \\ \phi_d = \frac{\partial v_q^*}{\partial \omega} \end{cases} \quad (5)$$

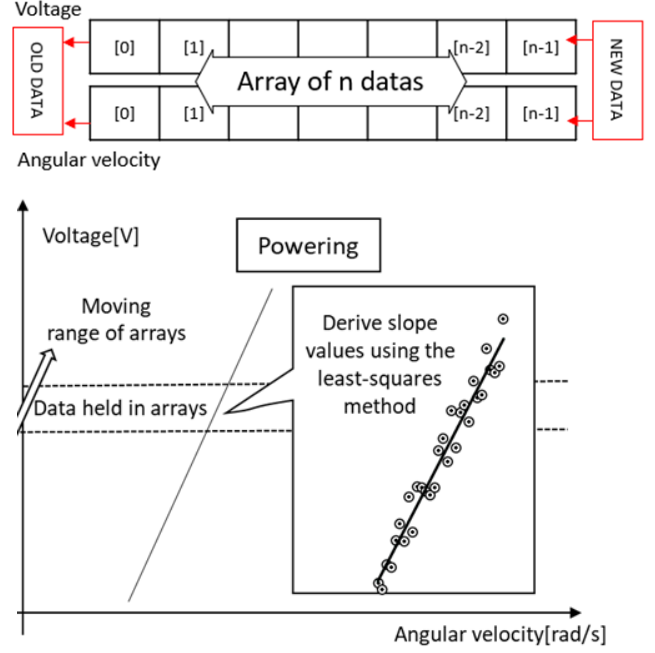


Fig. 4. Torque Estimation Algorithm

This estimation algorithm mirrors the current-type torque estimation formula in calculating torque through the outer product of the magnetic flux vector and the current vector. However, it diverges in that the magnetic flux is determined by partial differentiating the voltage during motor acceleration with respect to angular velocity, omitting the use of magnetic parameters in the flux calculation. The advantage of this torque estimation method lies in the differentiation of voltage by angular velocity, unaffected by voltage drops unrelated to angular velocity. Consequently, magnetic flux can be computed solely from the induced voltage. This approach facilitates magnetic flux and steady-state torque estimation during motor operation, impervious to fluctuations in the motor's magnetic parameters and angular velocity.

The specific torque estimation method is depicted in Fig. 4. Initially, the voltage and angular velocity are sampled at a frequency synchronized with the PWM carrier and stored in an array. Subsequently, the slope derived from this array using the least-squares method represents the calculated magnetic flux.

The results of estimating the magnetic flux and torque using this method (an example) during motor operation are illustrated in Fig. 5. (In this instance, i_d : -2 [A], i_q : 5 [A], motor specifications are provided in Table 1 in Section 5). The significant fluctuation observed at the beginning of the estimation is attributed to speed variations resulting from the torque step input and the time needed to populate the data array. However, by averaging the values after the fluctuation has stabilized, a highly accurate torque estimate can be achieved.

3.2 Process for Creating Torque Estimation Approximation Equations⁽¹⁷⁾

In this method, a three-dimensional torque approximation equation is formulated with d-q-axis currents and the permanent magnet electromotive force constant as variables. This equation aims to compensate for variations in currents and magnetic parameters

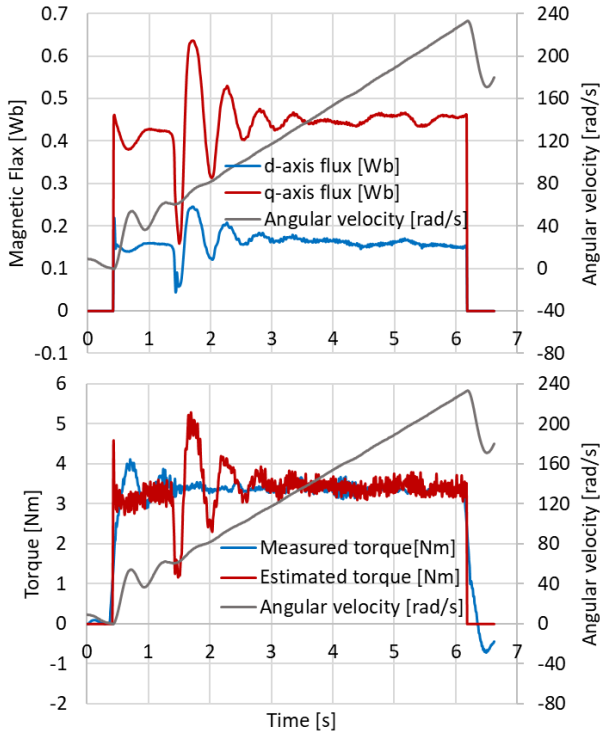


Fig. 5. Estimated torque and flux (proposed method)

Table 1. Specification of the PMSM and scale down model

Parameter	Value
DC link Voltage	300[V]
Pole	4
Rated Power	1.5[kW]
d-axis inductance	0.029[H]
q-axis inductance	0.087[H]
Winding resistance	1.35[Ω]
Electromotive force constant	0.234[Wb]
Rated rotation speed	1500[rpm]

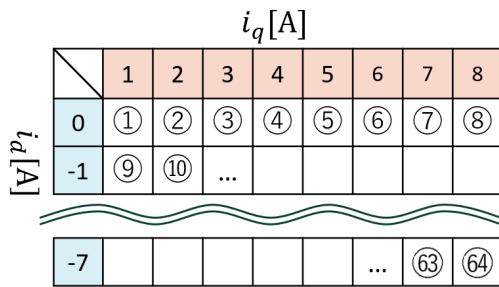


Fig. 6. Operation Schedule

resulting from rotor magnet temperature changes. The procedure is outlined below.

STEP1: Drive the motor with multiple current references and repeat acceleration and deceleration (zero current reference control) (Fig. 6). During acceleration, calculate the torque using the method described in Section 3.1. During deceleration (zero current reference control), only calculate the d-axis magnetic flux, and utilize this value as the permanent

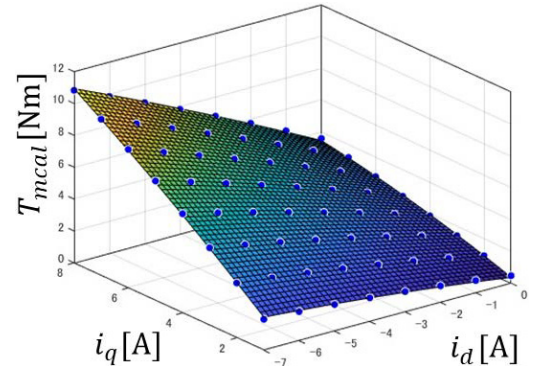


Fig. 7. Torque Approximation Surface

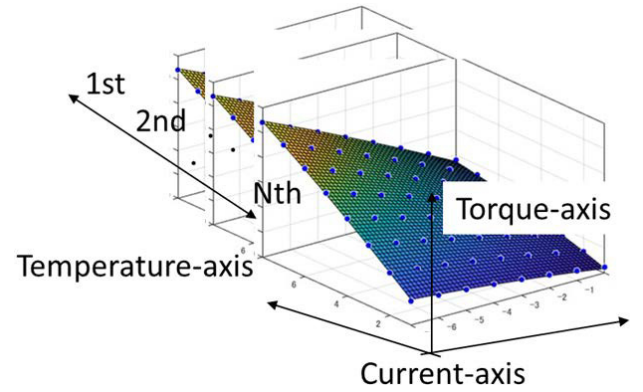


Fig. 8. Three-dimensional approximation of torque

magnet electromotive force constant.

STEP2: Utilizing the data acquired in STEP 1, conduct a least-squares approximation to approximate the torque to the second-order surface equation of the d-q-axis current, as illustrated in (6) and Fig. 7.

$$T_m(i_d, i_q) = p00 + (p10 * i_d) + (p01 * i_q) + (p20 * i_d^2) + (p11 * i_d * i_q) + (p02 * i_q^2) \dots \dots \dots (6)$$

$p00, p10, p01, p20, p11, p02$: Coefficients constituting the torque approximation formula

STEP3: Repeat STEP 1 and STEP 2 multiple times while incrementally increasing the magnet temperature. Consolidate the multiple second-order approximations obtained with the permanent magnet electromotive force constant at each temperature, as depicted in (7) and Fig. 8.

$$\begin{aligned} p00 &= a_{p00} \phi_f + b_{p00} & p10 &= a_{p10} \phi_f + b_{p10} \\ p01 &= a_{p01} \phi_f + b_{p01} & p11 &= a_{p11} \phi_f + b_{p11}, \dots \dots \dots (7) \\ p20 &= a_{p20} \phi_f + b_{p20} & p02 &= a_{p02} \phi_f + b_{p02} \end{aligned}$$

$a_{p00}, a_{p10}, a_{p01}, a_{p11}, a_{p20}, a_{p02}$: coefficient slope

$b_{p00}, b_{p10}, b_{p01}, b_{p11}, b_{p20}, b_{p02}$: coefficient intercept

The aforementioned sequence of operations generates a torque estimation approximation equation for each inductance, with each d-q-axis current and each magnet temperature (permanent magnet electromotive force constant) as the triple variables. When utilizing the torque estimation approximation equation, the permanent magnet electromotive force constant is obtained during motor coasting (with current kept at zero). Multiple second-order approximated dq current versus torque equations are created for each permanent magnet

electromotive force constant at different temperatures. By applying the obtained electromotive force constant, a d-q current versus torque approximation equation is established with the certain rotor temperature. Furthermore, by substituting the measured current into the approximated torque equation, PMSM torque can be estimated.

4. Torque Estimation and MTPA Angle Calculation using Approximation Equations

In the preceding chapter, a method was introduced for deriving a torque estimation approximation equation with d-q-axis current and magnet temperature (permanent magnet electromotive force constant) as the three variables. This formula is employed for both torque estimation and MTPA angle calculation, taking into account the magnetization status within the motor.

4.1 Torque Estimation using Approximation Equations In torque estimation, the estimated torque is determined by substituting the permanent magnet electromotive force constant and the d-q axis currents during motor operation, as described at the conclusion of the preceding chapter.

4.2 MTPA Angular Calculation using Approximation Equations⁽¹⁸⁾ Generally, the MTPA angle is derived by partial differentiating the torque equation with respect to the current phase angle and solving the extreme value problem, as depicted in (3). In our proposed approach, the torque equation, presented as (6), accounts for various flux variations. Consequently, by transforming (6) into polar coordinate form using the current vector length I_m^* and current phase angle θ in (8), we obtain the MTPA angle. Here, the current phase angle considers the d-axis current direction as 0 radians.

$$T_m(I_m, \theta) = p00 + (p10 * I_m^* \cos \theta) + (p01 * I_m^* \sin \theta) + (p20 * I_m^{*2} \cos^2 \theta) + (p11 * I_m^{*2} \sin \theta \cos \theta) + (p02 * I_m^{*2} \sin^2 \theta) \dots \dots \dots (8)$$

By partial differentiating Equation (8) with the current phase angle θ and setting it equal to zero, (9) is obtained.

$$\sin(2\theta + \beta) = \frac{-\sqrt{p10^2 + p01^2}}{I_m^* \sqrt{(p02 - p20)^2 + p11^2}} \sin(\theta + \alpha) \left(\alpha = \tan^{-1} \left(\frac{p01}{-p10} \right) \quad \beta = \tan^{-1} \left(\frac{p11}{p02 - p20} \right) \right) \dots \dots \dots (9)$$

Although it is not possible to solve for the current phase angle θ any further, the MTPA angle can be calculated by approximating the intersection of the right and left sides.

The principle of the approximate solution derivation is shown in Fig. 9. First, substituting the current vector length I_m^* into (9) determines the shapes of the two sine waves. The top of the sine waves and the zero-crossing point are connected by a straight line, and their intersection is obtained as an approximate solution.

5. Evaluation by Scale-down Model Experiments

The proposed method's accuracy in estimating torque and MTPA angle, along with torque control performance, is validated through a scale-down model experiment. While the primary aim of this paper is to enhance steady-state torque output performance, the experiments also confirm that the

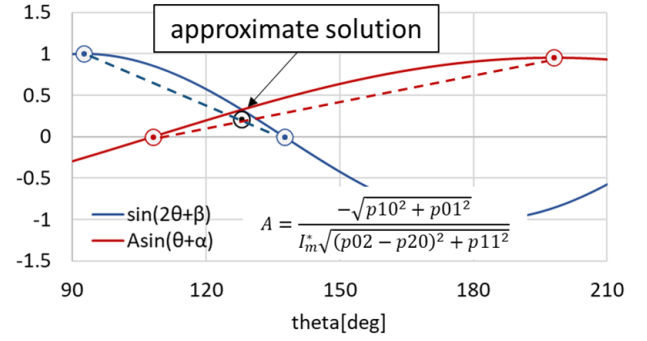


Fig. 9. MTPA approximate solution

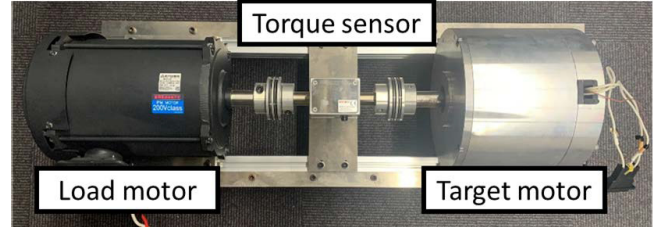


Fig. 10. Experimental Equipment

proposed control does not compromise torque dynamic performance. The appearance of the scale-down experimental system is depicted in Fig. 10. The specifications of the experimental 1.5 kW 4-pole IPMSM are detailed in Table 1. The scale-down model comprises the 4-pole experimental IPMSM connected to a 6-pole IPMSM rated at 1.5 kW as the load motor via a shaft equipped with a torque sensor.

5.1 Creation of Torque Estimation Approximation Equations

Using the experimental IPMSM, torque estimation approximation equations are generated using the proposed algorithm. In this paper, approximation equations are derived for the current phase angle ranging from 90° to 180°, where significant positive torques are observed. The operating points total 64, incremented by 1 A from 0 A to -7 A for the d-axis current and from 1 A to 8 A for the q-axis current. Acceleration and deceleration cycles are repeated between 0 and 230 rad/s to formulate the approximation equations in (7). The temperature axis consists of five points: 25°C, 50°C, 75°C, 100°C, and 125°C. The experimental IPMSM is operated at each temperature for 20 minutes while monitoring the temperature using thermocouples embedded in the stator. The permanent magnet electromotive force constant at each motor temperature is depicted in Fig. 11. The slope of the permanent magnet electromotive force constant aligns with the thermal constant of coercive force on the permanent magnet data sheet, indicating an actual increase in permanent magnet temperature⁽¹⁹⁾.

5.2 Verification of Torque Estimation Accuracy

The motor is operated solely using the current control system depicted in Fig. 2, and torque estimation accuracy at each temperature is assessed. The torque estimation methods outlined in equations (1) and (2), known as the current-type and voltage-type torque estimations, respectively, are evaluated and compared with the proposed method outlined in equation (6). The evaluation metric is the average error [%] between the actual torque measured by the torque sensor and the estimated torque across all operating points depicted in Fig. 6

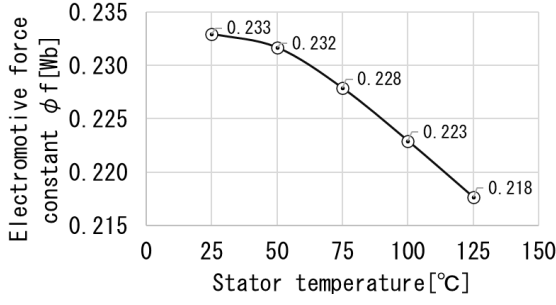


Fig. 11. Relationship between magnet flux and temperature

Table 2. Average error of measured torque and estimated torque

	25°C	50°C	75°C	100°C	125°C
Error with current Type (1) [Nm]	0.19	0.18	0.20	0.23	0.26
Error with voltage Type (2) [Nm]	0.35	0.36	0.40	0.44	0.49
Error with proposed method (6) [Nm]	0.12	0.11	0.11	0.12	0.11

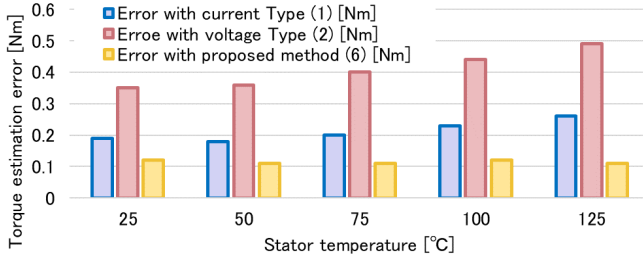


Fig. 12. Average error of measured torque and estimated torque

and all speed ranges. The verification results are presented in Table 2 and Fig. 12.

In the conventional method using equations (1) and (2), the torque estimation error increases as the temperature rises due to the effects of permanent magnet demagnetization and fluctuations in winding resistance. However, the proposed method exhibits higher torque estimation accuracy even at elevated temperatures.

The method of compensating for the variation of each parameter in equations (1) and (2) is also considered to be one means of obtaining high torque estimation results. However, the proposed method can perform torque estimation taking into account magnetic constant variations collectively, not only magnet flux variations due to temperature changes, but also inductance variations due to current and interference between the d-q axes. In this respect, it is also more advantageous than methods that compensate each parameter independently.

5.3 Verification of MTPA Angular Calculation Accuracy The accuracy of MTPA angle estimation by the proposed method is verified. The motor is driven using the torque feedback MTPA control system depicted in Fig. 1, and the torque reference value is incrementally raised from 2.0 Nm to 8.0 Nm. The proposed method is assessed by comparing it with the conventional MTPA theoretical equation (3). In this evaluation, the target motor is held in the medium speed range (around 130 [rad/s]) by the load motor to ensure that the evaluation is average over the entire speed range. The evaluation criterion is the percentage error with respect to the

Table 3. Accuracy of MTPA angle estimation (25°C)

Torque Reference [Nm]	2.0	4.0	6.0	8.0
I_m^* (Current vector length) [A]	3.4	5.6	7.5	9.1
θ_{MTPA} Analyzed value[deg]	118.25	123.38	125.83	127.21
θ_{MTPA} Proposed Approximation Method [deg]	119.30	124.09	126.31	127.65
θ_{MTPA} Conventional Method [deg]	117.95	123.38	125.79	127.19
Proposed Approximation Method Error [%]	0.89	0.58	0.38	0.35
Conventional Method Error [%]	0.26	0.00	0.03	0.01

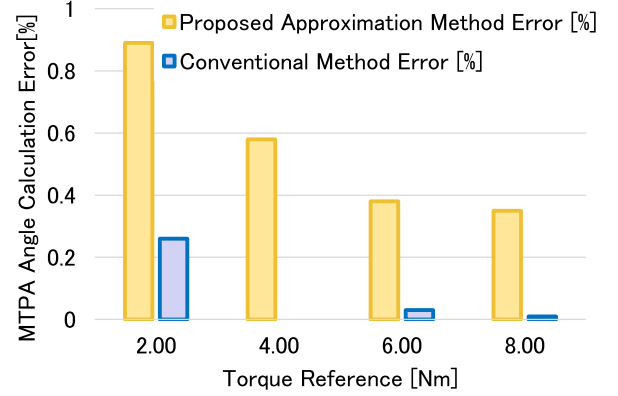


Fig. 13. Accuracy of MTPA angle estimation (25°C)

Table 4. Accuracy of MTPA angle estimation (125°C)

Torque Reference [Nm]	2.0	4.0	6.0	8.0
I_m^* (Current vector length) [A]	3.6	6.0	7.7	9.5
θ_{MTPA} Analyzed value[deg]	119.67	124.77	126.77	128.21
θ_{MTPA} Proposed Approximation Method [deg]	120.62	125.27	127.22	128.56
θ_{MTPA} Conventional Method [deg]	118.73	123.94	126.05	127.48
Proposed Approximation Method Error [%]	0.80	0.40	0.35	0.27
Conventional Method Error [%]	0.79	0.67	0.57	0.57

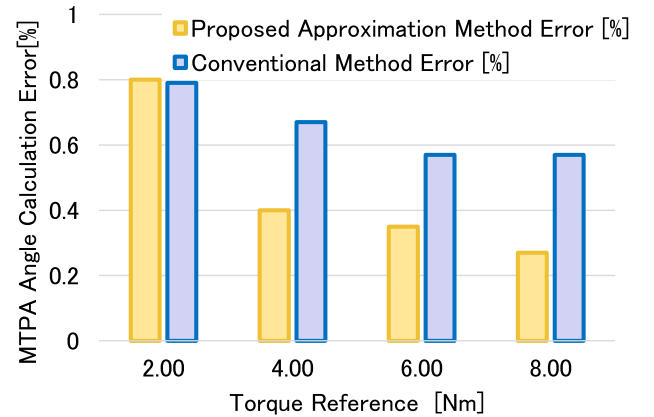


Fig. 14. Accuracy of MTPA angle estimation (125°C)

MTPA curve generated using the estimated torque. The estimated torque is derived using the method outlined in Section 3.1, as it provides a more precise measure of the electrical output torque.

Table 3 and Fig. 13 show the results at normal temperature (25°C), while Table 4 and Fig. 14 display the evaluation results at high temperature (125°C).

In the conventional method, MTPA angle calculation can be accurately performed within the normal temperature range

Table 5. Steady-state torque Control performance

		Output torque [Nm]		
Torque Reference [Nm]		2.00	5.00	8.00
25[°C]	Proposed	1.86	4.99	8.15
	Conventional	1.84	4.84	7.79
75[°C]	Proposed	1.89	5.01	8.11
	Conventional	1.82	4.82	7.73
125[°C]	Proposed	1.89	5.00	8.12
	Conventional	1.79	4.75	7.63

(25°C) since there is no variation in the permanent magnet electromotive force constant inside the motor. However, in the high temperature range (125°C), estimation accuracy deteriorates due to the variation in the electromotive force constant.

On the other hand, the proposed method can calculate the MTPA angle even at high temperatures, accounting for parameter fluctuations, thus maintaining estimation accuracy similar to that at normal temperatures. Additionally, unlike the conventional method, the proposed method's error remains consistent regardless of temperature, making it easier to compensate for calculation errors.

5.4 Verification of Steady-state Torque Control Performance The steady-state torque control performance of the proposed method is verified by driving the motor with the torque feedback MTPA control system shown in Fig. 1. The torque reference remains constant, and the error between the reference value and the output torque in steady state is evaluated.

The experimental conditions consist of three points on the reference torque axis (2 Nm, 5 Nm, and 8 Nm) and three points on the temperature axis (25°C, 75°C, and 125°C), totaling nine points.

In this evaluation, the target motor is held in the medium speed range (around 130 [rad/s]) by the load motor to ensure that the evaluation is average over the entire speed range. The control system of the conventional method used for comparison uses the current-type torque estimation formula in (1) for torque estimation and the MTPA theoretical formula in (3) for MTPA angle estimation. Table 5 and Fig. 15. show the experimental results.

In the conventional method, torque error increases with higher current and temperature regions. However, the proposed method exhibits no increase in torque error along the temperature axis direction. A minor torque error is noted along the current axis direction, attributed to the operating point being in the approximation's end region. In the case that the torque reference is 1.0 Nm order, the proposed method can improve the torque estimation error with 0.01 Nm order from the 0.1 Nm order as the average value over all torque range by the conventional method. In addition, by means of the proposed method, the torque control error can be improved by the simple one order function with the torque reference (Fig. 16). This is thanks to the successful compensation of the error by the temperature change.

5.5 Evaluation of Torque Dynamics The motor was driven by the torque-feedback MTPA control shown in Fig. 1,

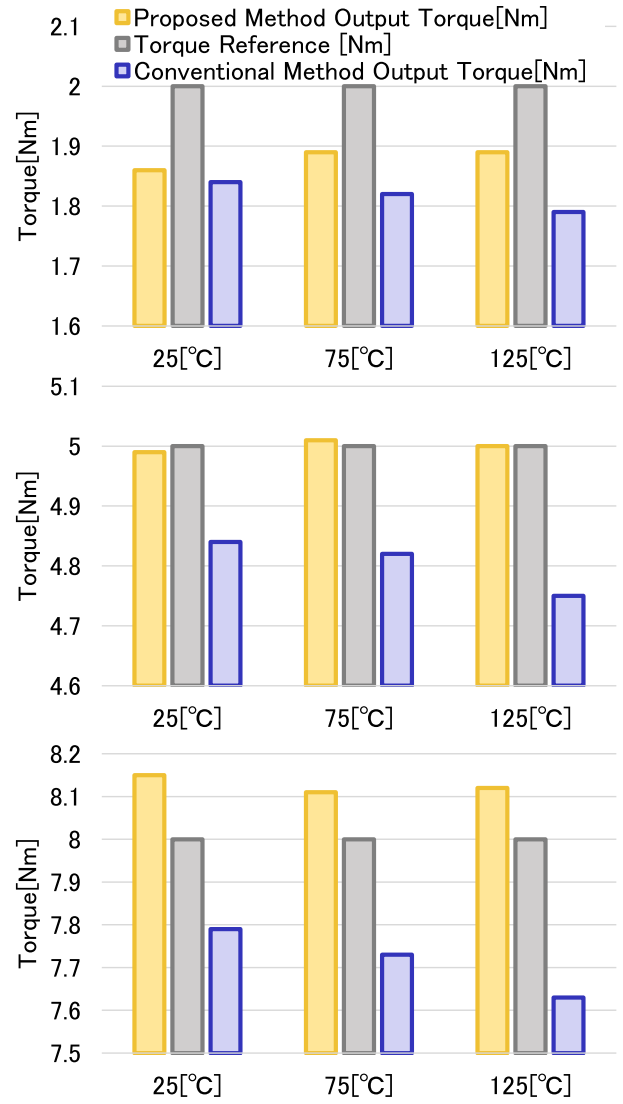


Fig. 15. Steady-state torque control performance

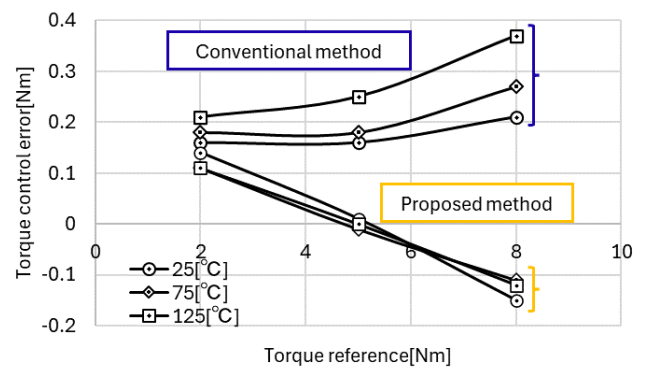


Fig. 16. Torque control error and stator temperature

and the frequency response from the torque reference value as an input to the actual torque as an output was evaluated. The torque reference T_m^* is fluctuated with an amplitude of 5% of the average reference torque, and the frequency of the fluctuation varies between 10 rad/s, 100 rad/s and 1000 rad/s. The cutoff frequency of the torque-feedback PI controller was designed to be 100 rad/s, which is identical to that of the current controller.

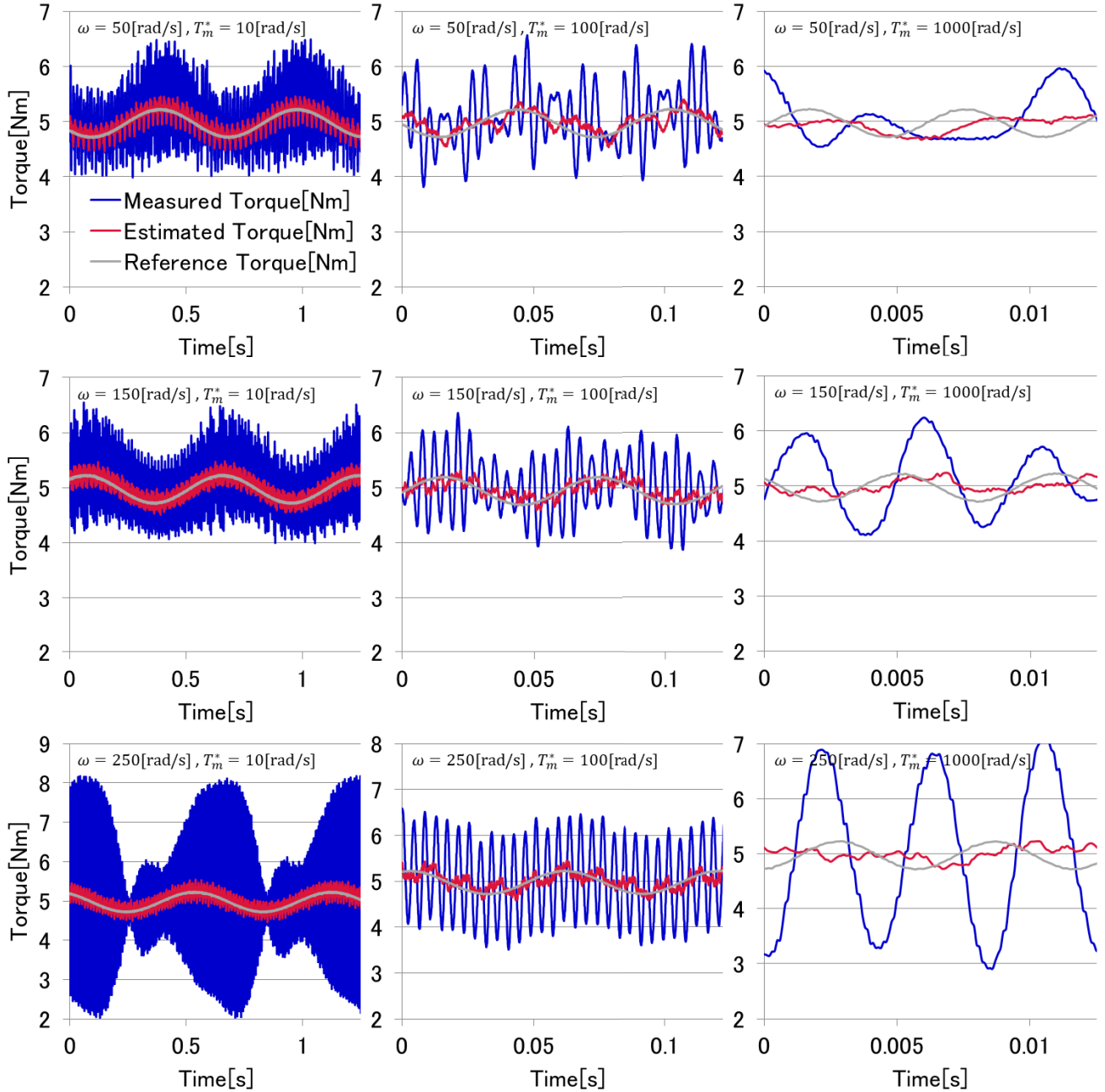


Fig. 17. Frequency response of output torque to Reference torque input

The motor's rotation speed during verification is maintained at low (50 rad/s), medium (150 rad/s), and high speeds (250 rad/s). The rated speed of this motor is 314 rad/s. The experimental results are depicted in Fig. 17. Though the frequency response at 1000 rad/s of the actual torque (blue waveform) cannot be precisely determined, the actual torque both at 100 rad/s and 10 rad/s does not have obvious delay against the torque reference. According to these results, it is revealed that the torque controller has the dynamic range with the 100 rad/s.

6. Conclusion

The inductance values of IPMSMs change depending on the current value, while the permanent magnet electromotive force constant changes depending on the rotor temperature. When these variations in magnetic parameters occur, the

controller cannot determine the minimum current required for the desired torque reference. As a result, there is an increase in copper losses and a deterioration in steady-state torque control performance.

Numerous methods have been proposed to compensate for fluctuations in magnetic parameters, but they tend to increase the complexity of controller calculations and analysis costs. Therefore, we propose a method that combines torque feedback MTPA control, known for its resilience to fluctuations in magnetic parameters like inductance and permanent magnet electromotive force constant, with an off-line obtained torque estimation approximation equation.

This enables steady-state torque control unaffected by fluctuations in magnetic parameters using a simple control block and straightforward offline data acquisition. Scale-down model experiments verify that a more accurate steady-state

torque control is achieved compared to the conventional method, without compromising dynamic torque control performance.

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Sota Kawashima (Member) received the M.E. degree from the School of Advanced Science and Engineering, Waseda University, Tokyo, Japan, in 2024. His research interests is high-efficiency control of permanent magnet synchronous motors for electric vehicle drives.



Keiichiro Kondo (Senior Member) received the B.S. and Ph.D. degrees in engineering from the Department of Electrical Engineering, School of Science Engineering, Waseda University, Tokyo, Japan, in 1991 and 2000, respectively. He joined the Railway Technical Research Institute, Kokubunji, Japan, in 1991 and was engaged in the research and development for power electronics applied to railway vehicle traction. From April 2007 to March 2018, he was with the Electrical and Electric Engineering Course, Graduate School and Faculty of Engineering, Chiba University, Chiba, Japan. Since April 2018, he has been a Professor with the Faculty of Science Engineering, Waseda University. His current research interests include power electronics, AC motor drives, energy storage devices, and wireless power transmission and their applications to railway systems. Prof. Kondo titled Professional Engineer Japan (Mechanical Engineering, Technical Management). He is a Member of IEEE.



Kazuhiko Matsunami (Member) received the M.Eng. degree from the Department of Electrical Engineering, the Graduate School of Shizuoka University, Japan in 1992. He joined Suzuki Motor Corporation in 1992 and is engaged in the development of electric powertrains, especially motor control.

